Guillem Hurault | Data Scientist

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ghurault • French • UK Settled Status

Data Scientist with a **PhD in Statistical Machine Learning** and an engineering degree. Deep expertise in **Bayesian modelling** and **time-series forecasting**. Experienced in managing research projects and delivering high-quality, maintainable, tested, and deployable ML solutions. Worked in research and industry in sports analytics, healthcare and energy.

Professional Experience

2025/12 - Now **Senior Data Scientist**, *Pythia Sports*, London, UK

- Lead the development of Bayesian models for a sports betting pipeline.

2026-01 - Now **Visiting Researcher**, *Imperial College London (Department of Bioengineering)*, UK, part-time

2024/08 - 2025/11 **Senior Data Scientist**, *CFP Energy*, London, UK

- Led a new data science team, designing end-to-end workflows, scalable software architecture and coding standards, which resulted in the production rollout of 10+ models used enterprise-wide.
- Develop models to forecast short- to long-term electricity demand, generation and prices; and provide valuations for Power Purchase Agreement (PPA) for renewable energy projects.

2024 - 2025 **Consultant - Bayesian Modelling**, *freelance*, London, UK, part-time

2022/09 - 2024/08 **Data Scientist**, *Pythia Sports*, London, UK

- Conducted statistical modelling research for a sports betting pipeline, incl.: Bayesian predictive modelling, bet optimisation under uncertainty, time-series forecasting of prices, 2D markerless pose estimation in videos.
- Managed and maintained the model for a complete betting market.
- Maintained, refactored and wrote technical documentation for our internal Python modelling packages.

2017/10 - 2022/09 *Imperial College London (Department of Bioengineering)*, UK

Research Associate (2022/06 - 2022/09) – Research Assistant (2017/10 - 2022/05)

- Analysed real-world and clinical trial patient data with state-of-the-art Bayesian models for time-series forecasting, to predict the evolution of eczema severity and generate personalised treatment recommendations.
- Published 10+ scientific articles to clinical and machine learning audiences. Developed software packages (EczemaPred, HuraultMisc). Presented work at international conferences. Reviewed several research papers.
- Supervised 25+ student projects, conducted interviews, managed internal packages, knowledge base and website.

2019 - 2021 **Teaching Assistant**, *Imperial College London (Department of Bioengineering)*, UK, part-time

- Probability & Statistics ○ Mathematics ○ Occasional teaching in Machine Learning and Brain Machine Interfaces.

2016/05 - 2016/07 **Research Intern**, *Laboratoire de Neurosciences Cognitives (CNRS UMR 7291), Aix-Marseille Université*, France

Analysed fMRI images using Machine Learning to understand the role of the oculomotor cortex in social perception.

Skills

Languages ○ Native **French** ○ Fluent **English** ○ Conversational Portuguese

Data Science ○ Machine Learning ○ Statistics ○ Bayesian modelling ○ Time-series forecasting ○ Uncertainty quantification ○ Visualisation ○ Missing values ○ Regularisation ○ Decision analysis ○ Clustering

Programming **Working knowledge**: ○ Python (incl. pandas, numpy, scikit-learn, scipy, ArviZ, darts, DeepLabCut) ○ R (incl. tidyverse, Rmarkdown/Quarto, Shiny) ○ Stan ○ LaTeX ○ RegEx ○ Git ○ SQL ○ bash

Basic knowledge: ○ Matlab ○ C# (incl. Infer.NET) ○ HTML ○ FFmpeg ○ Tableau

Software Engineering ○ Package development ○ Object-Oriented Programming ○ CI / CD ○ Testing ○ Refactoring ○ Technical documentation ○ Version control ○ MLOps

Technologies ○ Docker ○ Airflow ○ MLflow ○ GitHub & GitLab Actions ○ DataDog ○ Linux

Education

2018 - 2022 **PhD in Statistical Machine Learning**, *Imperial College London (Department of Bioengineering)*, UK
Thesis: Towards a data-driven personalised management of Atopic Dermatitis severity. Supervisor: Prof. Tanaka.

2016 - 2017 **MSc in Biomedical Engineering, Neurotechnology**, *Imperial College London*, UK, Distinction

2014 - 2018 **Master's Degree in Engineering**, *Ecole Centrale de Lyon*, one of France's top engineering schools

2014 - 2016 **Bachelor's Degree in Economics**, *Université Lyon 2*, France

Other Experiences

2019 - 2021 **Bioengineering PhD representative**, *Imperial College London*
Represented 200+ PhD researchers in departmental meetings, organised social and professional events.

2014 - 2017 **Engineering student**

- Investigated the evolution of eczema using Machine Learning methods in the Biological Control Systems Lab.
- Designed a genetic algorithm in a research project with LIRIS Lab (CNRS) to solve a scheduling problem.
- Supervised a 6-person team for HEXADRONE to design and test a security system to prevent a drone crash.

2015 **General Secretary**, *Forum Perspectives*
Organized a yearly career fair with 100 companies, 2000 students participating and a turnover of 250k€.

2015 **Treasurer, Communication coordinator and Editor** of Centrale Lyon's newspaper Piston Hebdo.

2015 Active committee member of Centrale Lyon's Cinema Society.

Selected Publications

- **G. Hurault**, J-F. Stalder, M. Saint Aroman, and R. J. Tanaka, "Data-driven personalised recommendations for eczema treatment using a Bayesian model of severity dynamics", *medRxiv*, 2024
- A. Duverdier*, **G. Hurault***, K. Thomas, A. Custovic and R. J. Tanaka, "Evaluation of measurement errors in the Patient-Oriented Eczema Measure (POEM) outcome", *Clinical & Experimental Allergy*, vol. 54, no. 3, p. 207-215, 2024
- **G. Hurault**, K. Pan, R. Mokhtari, B. Olabi, E. Earp, L. Steele, H. C. Williams and R. J. Tanaka, "Detecting eczema areas in digital images: an impossible task?", *JID Innovations*, vol. 2, no. 5, p. 100133, 2022
- **G. Hurault**, J-F. Stalder, S. Mery, A. Delarue, M. Saint Aroman, G. Josse and R. J. Tanaka, "EczemaPred: A computational framework for personalised prediction of eczema severity dynamics", *Clinical and Translational Allergy*, vol. 12, no. 3, p. e12140, 2022.
- **G. Hurault**, E. Roekevisch, M.E. Schram, K. Szegedi, S. Kezic, M.A. Middelkamp-Hup, P.I. Spuls and R. J. Tanaka, "Can serum biomarkers predict the outcome of systemic immunosuppressive therapy in adult atopic dermatitis patients?", *Skin and Health Disease*, vol. 2, no. 1, p. e77, 2022.
- **G. Hurault**, V. Delorieux, Y-M. Kim, K. Ahn, H. C. Williams and R. J. Tanaka, "Impact of environmental factors in predicting daily severity scores of atopic dermatitis", *Clinical and Translational Allergy*, vol. 11, no. 2, 2021.
- K. Pan, **G. Hurault**, K. Arulkumaran, H. C. Williams and R. J. Tanaka, "EczemaNet: Automating Detection and Assessment of Atopic Dermatitis", *International Workshop on Machine Learning in Medical Imaging*, 2020.
- **G. Hurault**, E. Domínguez-Hüttinger, S. M. Langan, H. C. Williams and R. J. Tanaka, "Personalised prediction of daily eczema severity scores using a mechanistic machine learning model", *Clinical & Experimental Allergy*, vol. 50, no. 11, pp. 1258–1266, 2020.

Full list available on my website.